

Disentangling the Performer from the Performance in Gendered Evaluation

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Abstract

We re-render professional tennis points with motion-capture animation, showing the identical performance with different performers. In our real-footage clips, women’s unforced errors are rated 0.29 points more severe than men’s, their winner points 0.11 less impressive. Two preregistered experiments (N = 3,730) decompose the gaps. Swapping the displayed gender on a synthetic performer makes no detectable difference in either rating. The re-rendering that makes the swap possible is itself a treatment: every re-rendered condition removes the extra severity of women’s errors, including the one that preserves the displayed gender, and the winner-points gap widens toward the larger gap real presentation had masked. Both real-presentation components exceed the tightest equivalence bounds on the swap; the displayed gender thus accounts for no statistically reliable part of either gap. Real presentation adds severity when women fail and applies a more lenient standard when they succeed; blinding lifts the penalty and withdraws the subsidy together, and correspondence designs that randomize a label alone detect neither.

Keywords: gender; discrimination; performance evaluation; blinding; online experiment; synthetic media

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1 Introduction

Evaluations of work are also evaluations of whoever appears to have done it. Would the same success be praised, and the same mistake blamed, as readily if a woman appeared to be responsible? The question is old, but the comparison it calls for does not occur in the field: no two real performances are identical, and evaluations of real people entangle what was done with who appears to have done it, with reputation and context. This paper constructs the comparison in professional tennis, a setting where men and women do the same job under the same rules. We re-render tennis points with motion-capture animation so that the identical performance can be shown as performed by its original players, by avatars of either gender, or by a robot, and we ask whether the same performance is judged differently when displayed as a man's and as a woman's.

Economics measures discrimination in evaluation in two main ways. Field studies compare outcomes for real people, with sports a favored setting because performance is public and unusually completely recorded for every worker ([Kahn, 2000](#)): professional basketball players are called for fewer fouls when the refereeing crew shares their race ([Price and Wolfers, 2010](#)), and baseball umpires favor pitchers of their own race or ethnicity on judgment calls, a bias that appears mainly where scrutiny is low ([Parsons et al., 2011](#)). Identification there comes from variation in evaluator-performer matchups; the evaluated performance itself is, at best, conditioned on, never held constant: playing style can be proxied, never fixed. Correspondence experiments remove that confound by construction: they randomize a group label on an otherwise fixed application, and the label alone moves outcomes, with resumes bearing White-sounding names drawing 50 percent more callbacks than equivalent resumes bearing African-American-sounding names ([Bertrand and Mullainathan, 2004](#)); the paradigm now spans dozens of studies ([Bertrand and Duflo, 2017](#)). Yet a label does not sit inertly beside the credentials it accompanies: the same degree, the same employer, the same gap in a work history can be read differently once the applicant's group changes. What a callback registers is therefore the label together with the reading it induces, and randomizing the label on a fixed application cannot separate the two.

The nearest evidence manipulates the cue on real footage. Blurring every human figure in

broadcast soccer clips, so that viewers cannot tell whether they are watching men or women, closes the rating gap between men’s and women’s highlight plays: 0.17 points on a five-point scale with gender visible, near zero with it concealed ([Gomez-Gonzalez et al., 2024](#)). The cue has to be visible for the gap to appear. Re-rendering asks the complementary question. Instead of hiding the cue it sets it, so the same play can be shown as a man’s and as a woman’s, and the performer who delivers it can be varied independently. Related manipulations place a displayed category on synthetic faces ([Reina et al., 2026](#)) or recolor real players ([Nessler et al., 2025](#)); neither holds a performance fixed while changing who delivers it. The nearest null comes from scripted delivery: when the same two-minute talk is presented by AI avatars or real actors of either gender, evaluations do not differ ([Franzen and Wöhner, 2026](#)). That design fixes content by script and finds that the presenter’s gender carries nothing; it leaves no outcome at stake in the judged performance and has no origin gap to decompose. Our design begins with both.

We build that design from the 2023 Wimbledon Championships. From eight matches, four men’s and four women’s, we select one point ending in a winner and one ending in an unforced error per match, as classified in the tournament’s official statistics, and re-render each point frame by frame with motion-capture animation, holding the performance fixed. Each point can then be shown in one of five conditions: the real footage, an avatar matched to the original player’s gender (the same-gender condition), a gender-neutral robot, or a single male or a single female avatar for all clips (the two fixed-gender conditions). In two preregistered experiments, 3,730 participants each viewed the eight clips of one experiment in one randomly assigned presentation and rated winner points for impressiveness (Experiment 1) or unforced errors for severity (Experiment 2); clip origin, men’s versus women’s tennis, varied within participant. The two fixed-gender conditions show identical content dressed as a man or as a woman, so a rating difference between them isolates the displayed gender. Tennis makes this possible where most work does not: its output is discrete, publicly observed, and officially classified, while most workplace output is heterogeneous and entangled with context. We measure a discretionary judgment. Judgments of that kind decide referrals, dismissal and rehiring in jobs where output is visible but the rating is left to a person

(Sarsons, 2022; Egan et al., 2022).

Three results emerge. In the real footage, both gaps are present and both run against women-origin clips: their unforced errors are rated 0.29 points more severe than men-origin errors, and their winner points 0.11 less impressive. Swapping the displayed gender on the synthetic performer moves neither rating at the 5 percent level; two estimators of the display effect never significantly disagree, and the tightest equivalence bounds are 0.12 points for winner points and 0.09 for unforced errors, below the components that do move. The third result is that the re-rendering is itself a treatment: removing the real performer moves both gaps. Every re-rendering reduces the extra severity of women-origin errors by 0.15 to 0.20 points, including the one that preserves the original player’s gender, and the four reductions are statistically indistinguishable. Re-rendering widens the winner-points gap instead, toward the larger gap that real presentation had masked. In the accounting of Section 3.4, real presentation carries about half of the observed severity gap and masks about half of the winner-points gap that identical content produces under a shared synthetic display; the displayed gender accounts for no statistically reliable part of either.

This answers the question the design was built around: whatever separates the judgment of men’s and women’s performances here, the displayed gender is no detectable part of it. What moves both gaps is real presentation, and it moves them in opposite directions. When women succeed it works in their favor, offsetting part of the gap the plays themselves carry: the signature of a lower expected standard, against which the same winner counts for more (Biernat and Vescio, 2002). When women fail it works against them, adding a severity that re-rendering removes. Blinding the evaluator to the performer therefore lifts a penalty and withdraws a subsidy at the same time. Section 4 develops the constraints: the opposite welfare signs reject any uniform attitude toward women; conditional on conventional beliefs about the women’s game, the extra severity runs against accurate statistical discrimination; and taste and inaccurate beliefs remain observationally equivalent without further data (Bohren et al., 2025a). The extra severity also shares its shape with the failure penalties documented for financial advisers and surgeons (Abel et al., 2024; Egan et al., 2022; Sarsons, 2022), and our design adds what those settings cannot vary: the same failure, shown with the

same displayed gender, is judged less harshly once the performer is replaced.

The findings contribute along three lines. For the measurement of discrimination, they add a manipulation the standard designs do not have: what varies is who performs, not the label attached to a fixed dossier. Randomizing the label alone would have found almost nothing here, and would have missed the two components that do move. The same lesson appears in [Bohren, Hull, and Imas \(2025b\)](#): applicants identical up to name and pronouns are evaluated indistinguishably, while the gendered recommendation letters they carry account for essentially the whole gap. Their unreachable component is produced at an earlier decision and recovered by re-running it; ours sits inside a single evaluation, and re-rendering recovers it wherever evaluation concerns recorded performance. For evidence on gendered evaluation, the two components are the finding: real presentation adds severity to women’s failures and softens the judgment of their successes, and a design that only conceals the cue cannot see either, because concealment leaves the performer in place. For blinding policy, the decomposition explains why field results disagree in sign. Field blinding conceals who produced the work and leaves the work in place: a screen at an audition, a name removed from an application, an anonymously graded script. Those interventions have not moved outcomes in the same direction ([Goldin and Rouse, 2000](#); [Behaghel et al., 2015](#); [Lavy, 2008](#)). Re-rendering replaces the performer instead, which is what separates the part of an evaluation concealment can reach from the part it cannot; Section 5 draws that out.

Section 2 describes the stimuli, conditions, and empirical strategy. Section 3 reports the results. Section 4 interprets them through the standard mechanisms of discrimination and the channels the data can bound, and Section 5 concludes.

2 Experimental Design

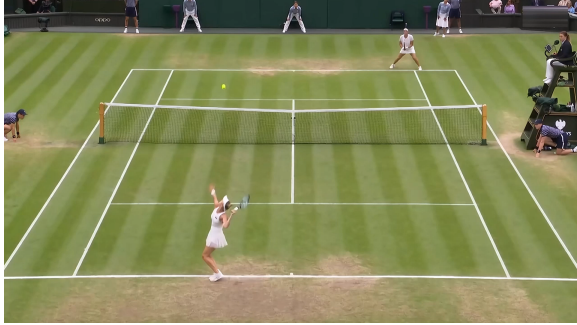
We conducted two preregistered experiments in which online participants rated video recordings of professional tennis points. Experiment 1 concerns winner points, rated for impressiveness; Experiment 2 concerns unforced errors, rated for severity. The two experiments share the same setting,

stimulus pipeline, and design, and differ only in the clips and the rating item. Re-rendering the same points with computer-generated performers lets us vary who appears to perform a play while holding the play itself fixed.

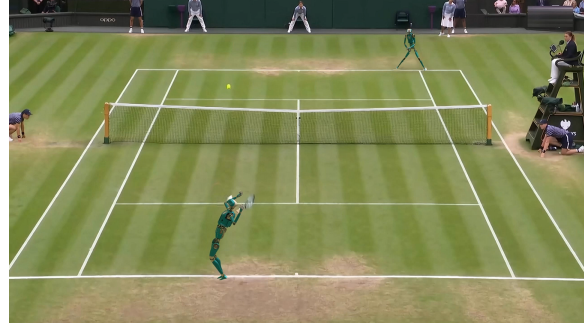
2.1 Setting and Stimuli

Professional tennis offers two properties that the design exploits. First, every point ends in one of three officially recorded ways: a *winner* (a shot the opponent cannot return), an *unforced error* (an errant shot attributable to the player’s own execution rather than the opponent’s play), or a forced error. These outcomes are defined and recorded by the same criteria in men’s and women’s tennis, so successes and failures are classified identically for both groups. Second, points are short, self-contained, and filmed from standardized angles, so a single point can be re-rendered frame by frame with the performance held fixed.

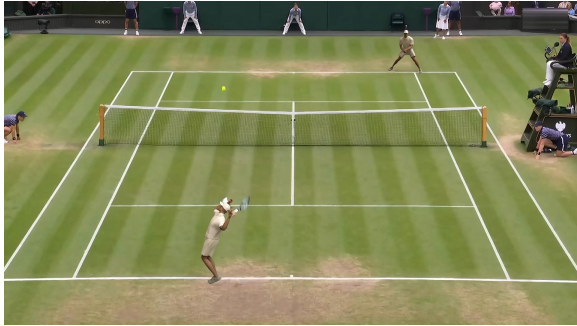
We drew stimuli from the 2023 Wimbledon Championships, quarterfinals onward, on the two main courts. From each of eight matches (four men’s, four women’s) we selected one point ending in a winner (Experiment 1) and one ending in an unforced error (Experiment 2), classified by the tournament’s official point statistics. The clips come from official highlight footage with standardized camera angles, and the men’s and women’s sets were matched on rally count and duration. We designed and modeled two humanoid avatar characters ourselves (one male, one female; Blender 4.1), each in five outfit variants matching the on-court attire; a third, gender-neutral robot character completes the synthetic set. We re-rendered each point with Wonder Studio (Wonder Dynamics), an AI-assisted visual-effects tool that performs markerless motion capture on the original video and composites the imported characters into the original broadcast scene, leaving the court, crowd, and camera view unchanged. Movement, ball trajectory, court position, and outcome were held fixed, so that the same play is performed by a real player, a robot, a male avatar, or a female avatar; Figure 1 shows the four appearances of one point, with the avatars’ genders legible in the composited scenes (panels c and d). All clips, in every condition, were presented without sound.



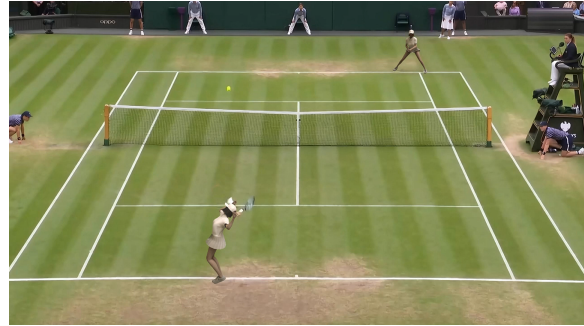
(a) Real footage



(b) Robot



(c) Male avatar



(d) Female avatar

FIGURE 1
THE SAME POINT ACROSS THE FOUR RENDERINGS

Note.—Stills from one stimulus point; the original player in this example is a woman. The re-rendered versions preserve the play of the point; only the performer's appearance changes. The same-gender condition reuses the avatars in (c) and (d), matched to each original player's gender. Panels (b)–(d) were produced with an AI-assisted visual-effects tool (Wonder Studio); the male and female avatar characters were designed by the authors (Section 2.1).

2.2 Experimental Conditions

Each participant was randomly assigned to one of five conditions that differ only in who appears to perform the eight plays. In the *real-footage* condition, participants watched the original broadcast clips of the four male and four female players. In the *same-gender avatar* condition, each play was re-rendered with the avatar of the original player’s gender (men’s points on the male avatar, women’s points on the female avatar), preserving the gender-origin pairing of the real footage while removing the real performer. In the *robot* condition, all eight plays were re-rendered with a robot character designed to be gender-neutral. In the *all-male* and *all-female avatar* conditions, all eight plays were re-rendered with the male avatar or with the female avatar, so that the two conditions differ only in the displayed gender. Comparing the two fixed-gender conditions isolates the displayed gender; comparing re-rendered conditions with real footage isolates what real presentation carries. Participants were told nothing about the clips’ provenance (Appendix A); the re-rendered performers are animated characters composited into the broadcast scene (Figure 1, panels b through d).

2.3 Outcome Measures

After each clip, participants answered a single rating item. In Experiment 1 the item was “How impressive is the win of this point?”, with response options displayed from *Very Impressive* to *Very Unimpressive*; in Experiment 2 it was “How severe is this error?”, displayed from *Very Severe* to *Very Minor*. Responses are coded 1–7 so that higher values indicate a more impressive win or a more severe error. Ratings were not incentivized for accuracy: impressiveness and severity have no objective standard against which a judgment could be scored. Stated evaluations are the object of interest here rather than a proxy for one, since what media markets aggregate is how audiences judge athletes. The design limits the concern that the measure is uninformative: the comparison is within participant and across randomly assigned presentations, so any general tendency to rate high or low is differenced out, and differences across participants in how widely they use the scale are balanced by randomization and addressed by the compression-adjusted specification (Table B.1).

In the one direct comparison we know of, male evaluators judging a woman’s failure, the stated measure ran, if anything, in the direction more favorable to women; [Abel et al. \(2024\)](#) read that divergence as a warning that unincentivized measures understate discrimination, which would make the real-footage gaps we estimate conservative. That argument bounds the gaps, not the nulls; for those, the equivalence bounds of Section 3.2 are the answer. Age, sex, and race were taken from Prolific profiles; education, household income, and tennis viewing frequency (from *Never* to *Daily*) were self-reported in the study. Throughout, displayed gender refers to the gender conveyed by the presentation (the real player or the avatar shown); rater sex refers to the sex from those profiles; clip origin refers to whether the point was played in the men’s or the women’s draw.

2.4 Samples and Procedure

Participants were recruited on Prolific, which provided a representative United States sample stratified by sex, age, and race; they were paid £0.50 for the approximately three-minute study. Data for both experiments were collected in July and August 2024; both studies received institutional review board approval and informed consent (see the Ethics and Preregistration statement). In Experiment 1, of 2,004 completed sessions, 1,997 matched Prolific records, 1,986 were approved submissions, and 1,964 reported watching all eight videos to completion. That item is self-reported; a per-clip playback flag records incomplete playback for a further 169 participants in Experiment 1 and 176 in Experiment 2, and excluding those as well leaves the interaction at -0.084 ($p = 0.061$) and -0.180 ($p < 0.001$) (Table B.1). We additionally excluded 67 participants who gave an identical rating to all eight clips (straight-line responders). The screens retain participants who saw the clips they rated and distinguished among them; results are substantively unchanged when the straight-line responders are retained (Table B.1). The analysis sample is $N = 1,897$ (944 female, 953 male; mean age 43.0 years, standard deviation 15.2), contributing 15,176 ratings.

Experiment 2 followed the same completion screening: of 2,014 completed sessions, 1,958 passed it. Recruitment on Prolific was configured to exclude Experiment 1 participants; twenty-five nonetheless took part in both, and are excluded under that recruitment criterion, leaving 1,933.

Excluding 100 participants who gave an identical rating to all eight clips leaves $N = 1,833$ (925 female, 908 male; mean age 45.3 years, standard deviation 15.8), contributing 14,664 ratings. Results with the 25 Experiment 1 participants retained are substantively unchanged (Table B.1). None of the sample restrictions described in this section was specified in the preregistration. Table 1 reports participant characteristics by condition. Randomization checks are consistent with chance imbalance: joint tests of the Table 1 characteristics across conditions, from multinomial logits of condition on all six characteristics, give $p = 0.176$ (Experiment 1) and $p = 0.643$ (Experiment 2). The only characteristic that differs significantly across conditions is household income in Experiment 1’s robot condition ($p = 0.003$, one of twelve single tests). Covariate-adjusted estimates appear in columns (2) and (4) of Tables 2 and 3; they move no interaction by more than 0.007.

The preregistered power analysis targeted approximately 400 participants per condition, providing 90 percent power to detect an interaction of $f^2 = 0.032$ at $\alpha = 0.01$. Sensitivity analyses on the final samples, at conventional benchmarks ($\alpha = 0.05$, two-sided, participant-clustered), indicate 80 percent power to detect an origin $\times$ re-rendering interaction of $b = 0.118$ (Experiment 1) or $b = 0.131$ (Experiment 2) on the 1–7 scale, and a direct male- versus female-avatar contrast of $b = 0.197$ or $b = 0.224$.

Every participant saw all eight clips of their experiment in an independently randomized order and rated each clip immediately after viewing it. Clips were introduced only as tennis highlights: no condition’s clips were described as replays of real points or as animations. Full survey wording, including the recruitment advertisement, consent excerpts, instructions, and all response options, is transcribed in Appendix A.

TABLE 1
SAMPLE CHARACTERISTICS BY CONDITION

	Real footage	Same-gender avatar	Robot	All-male avatar	All-female avatar
<i>Panel A: Experiment 1 (Winner Points)</i>					
Participants	396	347	388	381	385
Rating observations	3,168	2,776	3,104	3,048	3,080
Female (%)	49.0	53.9	47.9	46.2	52.2
Age (years)	42.5	43.2	42.4	43.0	43.9
White (%)	66.4	65.1	65.5	71.1	67.0
Tennis viewing frequency (1–7)	2.15	2.24	2.26	2.22	2.13
Years of education	15.0	15.1	15.2	15.0	15.1
Income ≥ USD 60,000 (%)	57.8	57.9	68.3	58.5	56.1
<i>Panel B: Experiment 2 (Unforced Errors)</i>					
Participants	353	386	354	396	344
Rating observations	2,824	3,088	2,832	3,168	2,752
Female (%)	54.7	53.1	50.3	48.0	46.2
Age (years)	44.9	44.8	46.1	44.8	46.0
White (%)	62.9	63.2	62.4	66.4	66.3
Tennis viewing frequency (1–7)	2.25	2.20	2.34	2.18	2.28
Years of education	15.2	15.3	15.2	15.1	15.2
Income ≥ USD 60,000 (%)	63.2	58.0	62.7	58.6	59.9

Note.—Participant characteristics by condition for the two analysis samples (straight-line responders excluded; the Experiment 2 sample additionally excludes 25 Experiment 1 participants per the recruitment protocol, Section 2.4). Experiment 1 elicited ratings of the impressiveness of winner points; Experiment 2 the severity of unforced errors. Age, sex, and race are Prolific platform variables; tennis viewing frequency (1 = never to 7 = daily), education, and household income are survey items (Appendix A). Years of education assign 12, 13, 14, 16, and 18 years to the five education categories of Appendix A. Rating observations count eight clips per participant.

2.5 Empirical Strategy

Let y_{ic} denote participant i 's rating of clip c , W_c indicate a women-origin clip, and $R_{a(i)}$ indicate assignment to a re-rendered condition. The pooled specification is

$$y_{ic} = \alpha + \gamma W_c + \beta R_{a(i)} + \delta (W_c \times R_{a(i)}) + \varepsilon_{ic}, \quad (1)$$

and the condition-specific specification replaces the re-rendering indicator with indicators $A_{a(i)}^k$ for the four re-rendered conditions,

$$y_{ic} = \alpha + \gamma W_c + \sum_k \beta_k A_{a(i)}^k + \sum_k \delta_k (W_c \times A_{a(i)}^k) + \varepsilon_{ic}. \quad (2)$$

All models are estimated by ordinary least squares with standard errors clustered by participant, the level at which presentation is randomized (Abadie et al., 2023); we report z statistics and exact p values (with more than 1,800 clusters per experiment, the normal reference distribution is appropriate). The coefficient γ is descriptive: it compares the four women's points with the four men's points in real footage, and therefore bundles the players' gender with everything else that differs between the two clip sets. The causal parameters are δ and the δ_k : presentation is randomized, so each is the experimental effect of that re-rendering on the origin gap.

Because every participant rates all eight clips, each participant's origin gap (the mean rating of their four women-origin clips minus the mean of their four men-origin clips) is the natural unit of analysis. The coefficients on W_c and its interactions in equations (1) and (2) are numerically identical to comparisons of mean participant-level gap scores across conditions, so every participant serves as their own control, and participant-clustered inference is equivalent to analyzing gap scores directly. The standard deviation of participant gap scores is 0.76 in Experiment 1 and 0.79 in Experiment 2, which we use to express effect sizes. The eight clips are the fixed universe of stimuli, and clip effects difference out of the gap score: adding clip fixed effects leaves every interaction estimate unchanged, and sensitivity to the stimulus set is assessed by leave-one-clip-out

estimates (Table B.1). Covariate-adjusted versions of both equations add the Table 1 characteristics, demeaned, as main effects and as interactions with W_c .

The displayed-gender test uses two estimators on the three human-avatar conditions (same-gender, all-male, all-female), both with clip fixed effects. The between-condition contrast compares ratings of identical content in the all-female and the all-male conditions. The within-participant estimate uses the same-gender condition: its participants see women-origin clips on the female avatar and men-origin clips on the male avatar, so any display effect enters their origin gap, and subtracting the fixed-gender conditions' content baseline isolates it. Under an additive model the two estimators coincide; Table 4 reports both, with a test of their equality. Equivalence testing follows the two-one-sided-test procedure (Lakens, 2017); we report the smallest symmetric margin the test rejects at the 5 percent level.

Three of the four condition interactions in equation (2) (the robot, all-male, and all-female conditions) were preregistered as non-directional hypotheses. The same-gender condition was registered as a manipulation check; its use as a focal contrast, the pooled δ , the displayed-gender tests of Table 4, and all equivalence bounds are post hoc.

3 Results

Throughout this section, parentheses report participant-clustered standard errors.

3.1 Gaps in Real Footage

In real footage, both gaps are present, and both run against women-origin clips. Women-origin winner points were rated less impressive than men-origin winner points: the control-condition gap is -0.105 (0.037), $p = 0.005$, or -0.14 gap-score standard deviations (Table 2).¹ Women-origin unforced errors were rated more severe than men-origin errors: 0.292 (0.042), $p < 0.001$, or 0.37

¹Exploratory splits locate this Experiment 1 gap among male raters (male -0.222 , female $+0.015$, interaction $p = 0.001$; not tabulated). The re-rendering interactions of Section 3.3 do not differ by rater sex (Table C.1): the level of the control-condition gap decouples from the components that move it.

gap-score standard deviations (Table 3). Because origin is a fixed property of the clips, these gaps describe the two clip sets as presented in real footage.

TABLE 2
EXPERIMENT 1 (WINNER POINTS): ORIGIN GAP AND RE-RENDERING

	(1) Pooled	(2) Pooled	(3) By condition	(4) By condition
Women-origin clip	−0.105*** (0.037)	−0.106*** (0.037)	−0.105*** (0.037)	−0.106*** (0.037)
× re-rendered (any condition)	−0.099** (0.042)	−0.099** (0.042)		
× same-gender avatar			−0.046 (0.056)	−0.051 (0.056)
× robot			−0.084 (0.054)	−0.080 (0.053)
× all-male avatar			−0.128** (0.054)	−0.121** (0.054)
× all-female avatar			−0.135*** (0.052)	−0.139*** (0.052)
Mean rating, men-origin clips	5.55	5.55	5.55	5.55
Covariate controls	No	Yes	No	Yes
Observations	15,176	15,176	15,176	15,176
Participants (clusters)	1,897	1,897	1,897	1,897

Note.—Ordinary least squares on ratings of the eight Experiment 1 clips (1 = very unimpressive to 7 = very impressive), all five conditions; standard errors clustered by participant in parentheses; p -values and stars are z -based. Women-origin clip = indicator for the four women’s points; its coefficient is the control-condition origin gap, and the interactions give the change in that gap under each re-rendering. The mean-rating row is the men-origin mean in the control condition. With every participant rating all eight clips, these coefficients equal comparisons of participant-level gap scores (mean rating of women-origin minus men-origin clips). Covariate controls are the Table 1 characteristics, demeaned, entered as main effects and as interactions with the women-origin indicator. Condition-level main effects are included but not shown. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

TABLE 3
EXPERIMENT 2 (UNFORCED ERRORS): ORIGIN GAP AND RE-RENDERING

	(1) Pooled	(2) Pooled	(3) By condition	(4) By condition
Women-origin clip	0.292*** (0.042)	0.289*** (0.042)	0.292*** (0.042)	0.289*** (0.042)
× re-rendered (any condition)	−0.172*** (0.047)	−0.167*** (0.047)		
× same-gender avatar			−0.155*** (0.057)	−0.153*** (0.056)
× robot			−0.204*** (0.058)	−0.199*** (0.058)
× all-male avatar			−0.152*** (0.059)	−0.147** (0.059)
× all-female avatar			−0.180*** (0.061)	−0.173*** (0.061)
Mean rating, men-origin clips	4.58	4.58	4.58	4.58
Covariate controls	No	Yes	No	Yes
Observations	14,664	14,664	14,664	14,664
Participants (clusters)	1,833	1,833	1,833	1,833

Note.—Ordinary least squares on ratings of the eight Experiment 2 clips (1 = very minor to 7 = very severe), all five conditions; standard errors clustered by participant in parentheses; p -values and stars are z -based. Women-origin clip = indicator for the four women’s points; its coefficient is the control-condition origin gap, and the interactions give the change in that gap under each re-rendering. The mean-rating row is the men-origin mean in the control condition. With every participant rating all eight clips, these coefficients equal comparisons of participant-level gap scores (mean rating of women-origin minus men-origin clips). Covariate controls are the Table 1 characteristics, demeaned, entered as main effects and as interactions with the women-origin indicator. Condition-level main effects are included but not shown. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

3.2 Swapping the Displayed Gender

The sharpest test of a displayed-gender channel compares the two fixed-gender conditions: identical content dressed as a man or as a woman. Table 4 reports the two estimators of this display effect. For winner points, the between-condition contrast is 0.006 (0.070), $p = 0.927$, and equivalence testing rejects effects of ± 0.122 or larger; the within-participant implied estimate (Section 2.5) is 0.086 (0.050, $p = 0.086$), marginal at the 10 percent level and bounded at ± 0.168 . For unforced errors, the between-condition contrast is 0.148 (0.080, $p = 0.065$), marginally positive at the 10 percent level, and its bound of ± 0.279 cannot rule out a moderate severity effect of displaying a female performer on this estimator alone; the within-participant estimate, however, is 0.011 (0.049, $p = 0.829$), bounded at ± 0.091 , below the real-presentation components estimated in Section 3.4. The two estimators are not significantly different (equality $p = 0.360$ and $p = 0.146$), and the origin gap itself is indistinguishable between the two fixed-gender conditions (the difference between the all-female and all-male interactions in Tables 2 and 3: -0.007 , $p = 0.901$, for winner points; -0.028 , $p = 0.643$, for unforced errors). On a synthetic performer, neither display estimator reaches significance at the 5 percent level.

TABLE 4
DISPLAYED-GENDER TESTS ON IDENTICAL CONTENT

	Estimate	Standard error	90% confidence interval	Equivalence margin
<i>Experiment 1 (Winner Points)</i>				
Between-condition contrast	0.006	(0.070)	[−0.109, 0.122]	0.122
Within-participant implied estimate	0.086*	(0.050)	[0.003, 0.168]	0.168
Difference between the two estimates	−0.079	(0.087)	$p = 0.360$	
<i>Experiment 2 (Unforced Errors)</i>				
Between-condition contrast	0.148*	(0.080)	[0.016, 0.279]	0.279
Within-participant implied estimate	0.011	(0.049)	[−0.070, 0.091]	0.091
Difference between the two estimates	0.137	(0.094)	$p = 0.146$	

Note.—Ordinary least squares on ratings in the three re-rendered human-avatar conditions (same-gender, all-male, all-female) with clip fixed effects; standard errors clustered by participant; p -values and stars are z -based. Both rows estimate the effect of displaying a performance on a female rather than a male body. The between-condition contrast compares the all-female with the all-male condition on identical content, averaged over the two origins. The within-participant implied estimate is the origin gap in the same-gender condition minus the mean origin gap in the two fixed-gender conditions; under an additive display effect the two estimators target the same parameter. The equivalence margin is the smallest symmetric bound rejected by two one-sided tests at the 5 percent level. These tests were not preregistered. Samples: 8,904 ratings from 1,113 participants (Experiment 1) and 9,008 from 1,126 (Experiment 2). * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

3.3 Removing the Performer, Keeping the Gender

Removing the real performer collapses the extra severity attached to women-origin errors. Every re-rendering reduced the Experiment 2 gap: the four interactions range from -0.152 (0.059) to -0.204 (0.058) (Table 3, column (3)) and are statistically homogeneous, $\chi^2(3) = 1.09$, $p = 0.778$. The same-gender avatar, which preserves the performer’s apparent gender, reduced the gap as much as the conditions that remove or replace it: -0.155 (0.057), $p = 0.006$. Pooled across the four re-rendered conditions, the change is -0.172 (0.047), $p < 0.001$, or -0.22 gap-score standard deviations, leaving a residual gap of 0.121 (0.021), about 40 percent of the real-footage gap. A level decomposition puts most of the change on the women-origin clips: re-rendering lowered the rated severity of women-origin errors by 0.252 (0.072), $p < 0.001$, against a smaller and statistically insignificant change on men-origin errors (-0.081 , $p = 0.246$). Because re-rendered ratings in Experiment 2 were compressed toward the scale midpoint (shrinkage slope $\lambda = 0.70$) and the women-origin error clips sat farther from the midpoint, part of the raw change could be mechanical. With the re-rendering $\times$ clip-extremity control, the Experiment 2 change remains -0.102 (0.051), $p = 0.045$, and the compression-adjusted estimate pooled across the two experiments is -0.105 (0.033), $p = 0.001$ (Table B.1).

For winner points, the picture is weaker and runs in the same raw direction. Every re-rendered condition shows a numerically wider gap; the change is significant in the two fixed-gender conditions (-0.128 (0.054), $p = 0.018$; -0.135 (0.052), $p = 0.010$) but not in the same-gender or robot conditions ($p \geq 0.117$; Table 2, column (3)), and the four changes are again homogeneous, $\chi^2(3) = 3.25$, $p = 0.355$. Pooled across the four re-rendered conditions, the change is -0.099 (0.042), $p = 0.018$, or -0.13 gap-score standard deviations; compression is negligible in this experiment ($\lambda = 0.96$), and the adjusted estimate is -0.107 (0.043), $p = 0.013$. At eight clips, with origin a clip-level attribute, this adjustment is not uniquely determined in Experiment 1: equally flexible control bases that absorb similar shares of the clip-level variation return adjusted estimates between -0.138 and -0.020 (not tabulated), an identification limit of the eight-clip design rather than a failed check, and the compression-adjusted claim therefore rests on the estimate pooled across

the two experiments. Within the four re-rendered conditions taken together, the winner-points gap is -0.205 (0.020), $p < 0.001$: with the real players absent, these women-origin points drew lower ratings than the men-origin points (in the two fixed-gender conditions alone, the estimand Section 3.4 uses as the content baseline, it is -0.237). Re-rendering also lowered rating levels overall in Experiment 1 (men-origin clips -0.194 (0.056); women-origin clips -0.293 (0.055); both $p < 0.001$), a general cost of synthetic presentation. Figure 2 displays the ten condition-level gaps.

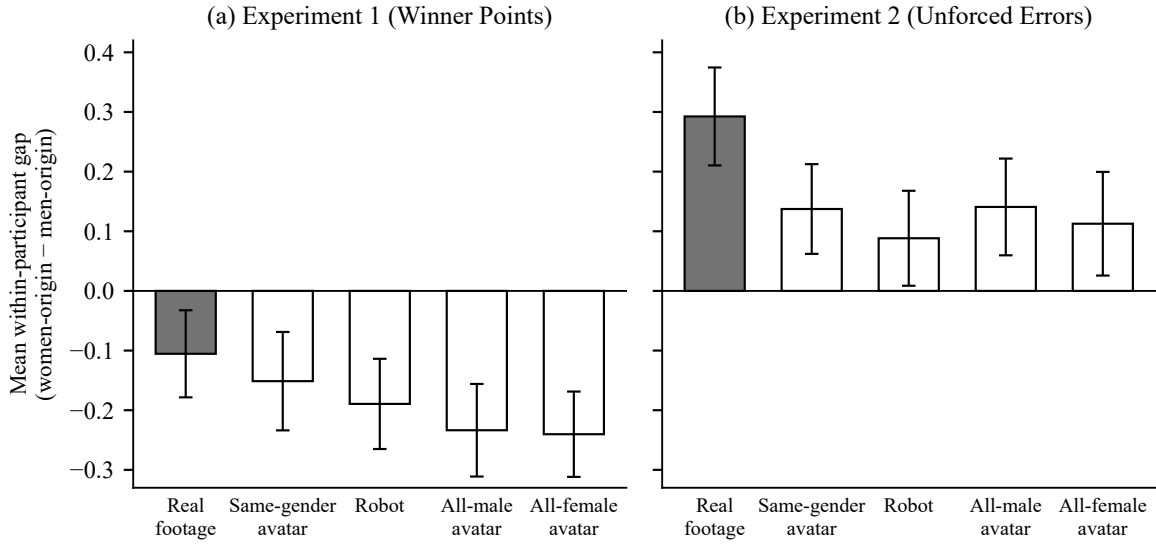


FIGURE 2
THE WITHIN-PARTICIPANT ORIGIN GAP BY CONDITION

Note.—Each bar is the mean participant-level gap score in that condition (the participant’s mean rating of women-origin clips minus men-origin clips); whiskers are 95% confidence intervals from participant-level standard errors. The solid bar is the real-footage condition; open bars are the re-rendered conditions. The two panels share the vertical scale. Sample sizes by condition as in Table 1.

3.4 Decomposition

The two experiments admit a single accounting. In each, the real-footage gap decomposes into the gap the same clips carry under a shared synthetic display (the content baseline, estimated in the fixed-gender conditions) plus what real presentation adds.² -0.105 combines a content baseline of

²The content baseline equals the real-footage gap plus the participant-weighted average of the two fixed-gender interaction coefficients in Tables 2 and 3; weighting the two conditions equally instead moves the components by at most 0.001.

-0.237 with a real-presentation component of 0.132 (0.046), $p = 0.004$, so real footage masks about half of the content baseline. For unforced errors, the observed gap of 0.292 combines a content baseline of 0.128 with a component of 0.165 (0.052), $p = 0.001$, so real presentation carries about half of the observed gap. The displayed-gender channel is too small to account for either component: in each experiment the tighter of the two equivalence bounds sits below it (± 0.122 against 0.132 for winner points, ± 0.091 against 0.165 for unforced errors), and the two estimators never significantly disagree (Section 3.2). The displayed gender thus accounts for no statistically reliable part of either gap.

On the raw rating scale the two effects share a sign: re-rendering lowers women-origin ratings relative to men-origin ratings for impressiveness and for severity alike. When the experiments are stacked, the two changes are statistically indistinguishable (difference -0.072 (0.063), $p = 0.250$), with a common re-rendering effect of -0.134 (0.031), $p < 0.001$, about a seventh of a point on the seven-point scales. An account in which re-rendering symmetrically reduces an anti-women distortion in both experiments would require opposite-signed raw changes and is rejected, $\chi^2(1) = 18.66$, $p < 0.001$. In welfare terms the components run in opposite directions: real presentation is favorable to women-origin winner points, masking part of a gap that would otherwise show, and unfavorable to women-origin errors, adding severity that no re-rendering retains.

3.5 Robustness

Table B.1 collects the specification battery for the re-rendering effect. The estimates are essentially unchanged when the 67 and 100 straight-line responders are retained (-0.095 , -0.161 , and -0.127 pooled across the two experiments), when the 25 Experiment 1 participants are retained in Experiment 2 (-0.099 , -0.175 , -0.136), and under the compression adjustment (-0.107 , -0.102 , -0.105 , the last at $p = 0.001$). Randomization inference (Young, 2019) gives permutation p -values of 0.018 , < 0.001 , and < 0.001 (Figure B.1). Leave-one-clip-out estimates of the compression-adjusted effect pooled across the two experiments are negative for all 16 clips, ranging from -0.137 to -0.064 with a worst-case $p = 0.058$. Within Experiment 2 alone, all eight leave-one-clip-out

estimates of the collapse remain significant at the 5 percent level (the weakest drop leaves -0.097 , $p = 0.039$); Experiment 1’s weaker change does not survive every drop (without clip 6: -0.037 , $p = 0.407$; not tabulated). Covariate adjustment in the main tables (columns (2) and (4) of Tables 2 and 3) moves no interaction by more than 0.007. The preregistered rater-sex moderations are null, and the remaining exploratory moderations and display-level contrasts appear in Table C.1.³

4 Mechanisms

The results split the gaps into parts: displayed gender moves no rating at the 5 percent level, and both real-footage gaps change when the real performer is removed. This section interprets that pattern with the standard mechanisms of discrimination, tests the narrower channels our data can reach, and states what remains unseparated.

4.1 The Displayed Category: Bounding the Classical Channels

Economics traditionally distinguishes taste-based discrimination, a preference against members of a group (Becker, 1957), from statistical discrimination, in which evaluators use a group attribute as a signal of unobserved quality (Phelps, 1972; Arrow, 1973). Our fixed-gender conditions confront both mechanisms with the same test: identical plays, dressed as a man or as a woman. A distaste keyed to the displayed female category would move ratings; so would an inference from the displayed category to the quality of the play. Neither does at the 5 percent level (Section 3.2). The null is not guaranteed by the synthetic performer. Evaluators apply gender stereotypes to machines readily: gendered computer voices draw gender-typed responses (Nass et al., 1997; Nass and Moon, 2000), and a displayed category on synthetic faces has moved evaluations (Reina et al., 2026); yet when scripted talks are delivered by avatars of either gender, ratings do not move (Franzen and

³Two Panel B contrasts in Table C.1 clear the 5 percent level and deserve note. The female-rater between-condition contrast (+0.255) has its within-participant twin directly below it at -0.012 (0.067): the two estimators of the same quantity part ways only in the noisier between-condition form, so we read the starred cell as between-arm noise. The all-female-versus-robot contrast (+0.192) is a level difference between display conditions, common to both clip origins, not an origin interaction; the gap claims of Sections 3.2 to 3.4 are unaffected.

Wöhner, 2026), and on our clips they do not detectably move either. A synthetic performer can carry a signal and can serve as a target; here it does neither, and the bound on how much it carries is estimated, not built into the design. What the test cannot do is apportion. Designs that split a gap between the two mechanisms shut one channel at a time while the person stays real, as when List (2004) varies whether dealers know that reservation values are randomly drawn and watches the group gap close; our swap moves both channels through the same displayed category, so it bounds their net rather than apportioning it, and exactly offsetting nonzero channels, though contrived, are not excluded (Bohren et al., 2025a). Within those limits the conclusion is one-directional: both classical mechanisms, in their displayed-gender versions, sit, on their tightest bounds, below the components that do move, and whatever generates those components must travel with real presentation.

4.2 An Outcome-Contingent Standard

What real presentation carries is not a uniform attitude toward women. An animus that depressed evaluations of real women across the board would make their winner points less impressive and their errors more severe; instead, real presentation is favorable on winner points and unfavorable on errors, and Section 3.4 rejects any single-signed distortion. What survives is outcome-contingent: an evaluative standard lenient toward real women’s successes and harsh toward their failures.

The lenient side reads naturally as shifting standards: subjective ratings are made relative to group reference points, so the same objective winner rates as more impressive against a lower expected standard (Biernat and Manis, 1994; Kobrynowicz and Biernat, 1997). That account indexes the reference point by the target’s perceived category. On subjective scales a shifted standard can offset in the rating, so the fixed-gender null alone cannot rule out a silently invoked standard (Biernat and Manis, 1994); what the design shows is that the masking appears only with the real players on screen, so the operative standard travels with real presentation, not with the category the display announces.

The harsh side runs against rational base-rate adjustment. Because re-rendering holds each play

fixed, the design supplies the ground truth an outcomes-based test needs (Bohren et al., 2025a). If raters believe unforced errors to be at least as common in the women’s game, the direction gender beliefs typically take in male-typed domains (Bordalo et al., 2019), then a woman’s error is less surprising and should, if anything, be judged less severely. We observe the opposite. Conditional on that belief direction, the severity component is inconsistent with accurate statistical discrimination. It remains consistent with two sources our data cannot separate: a distaste specific to women’s failures, or an inaccurate belief that reads a woman’s error as especially diagnostic of low ability, a stereotype in the sense of Bordalo et al. (2016). The two are observationally equivalent without further data (Bohren et al., 2025a). Experiments can break this equivalence by construction: Barron et al. (2025) shut the taste channel by making hiring choices inconsequential to the candidates, so that with true productivity equal, the residual gap identifies belief-based discrimination. The field counterpart works from the belief side: Montoya et al. (2026) give Chilean loan officers the banking regulator’s statistics on women’s repayment; the minority whose priors ran the other way update them and discrimination in that subgroup tends to attenuate, though the authors caution that none of those coefficients survives their multiple-hypothesis corrections and read the minority as too small a share of the market for the overall approval gap to move. Our design fixes the ground truth but leaves both channels open; eliciting raters’ beliefs about group error rates alongside the ratings, an inexpensive addition, would separate them.

The harsh side also has an external anchor. Across financial advice, adviser misconduct, and surgical referrals, evaluators penalize women more than men for the same failure (Abel et al., 2024; Egan et al., 2022; Sarsons, 2022), and the gap lives only there: nothing comparable appears before outcomes, where advisor gender moves nothing (Abel et al., 2024) and turnover absent misconduct is identical by gender (Egan et al., 2022), or after successes (Abel et al., 2024; Sarsons, 2022). Evaluations that stake no outcome at all fit the same pattern: scripted presentations delivered by real actors of either gender draw indistinguishable ratings (Franzen and Wöhner, 2026). In all three field settings a real woman stands behind the evaluated output; in Abel et al. (2024) she is represented to participants by a demographically matched portrait. In ours the performance is equally real, and

only the performer is replaced. The extra penalty is present throughout those settings and absent once the performer is replaced, and within our design it goes whenever the real performer does, even with the displayed gender kept. What the penalty tracks is therefore real presentation, not the gender label the display carries. One tension is worth flagging: the two settings that split the penalty by evaluator gender find it among men (Abel et al., 2024; Egan et al., 2022), though the third finds no significant difference by evaluator gender (Sarsons, 2022), while our preregistered rater-sex moderations are null (Table C.1); we read our evidence as uninformative on that margin rather than discordant.

4.3 What Re-rendering Bundles: Limits of the Design

This is where the design reaches its limit. Real presentation bundles the belief that a real woman stands behind the play with the visual richness of broadcast footage, the faces, reactions, and context that no re-rendering retains. Every condition that removes one removes the other, so the components of Section 3.4 belong to the bundle.

Narrower channels are testable, and null. Attention quantity, the margin on which group cues shift how much scrutiny an evaluator applies (Bartoš et al., 2016), shows no trace: median time on a clip page is 22 to 24 seconds in every cell; the origin $\times$ re-rendering interaction on log viewing time is -0.009 ($p = 0.558$) in Experiment 1 and -0.001 ($p = 0.972$) in Experiment 2; and controlling for clip position and viewing time moves the estimates by less than 0.002 (not tabulated). Equal seconds can still hide different processing, so this bounds duration, not depth. Familiarity with the players is similarly flat: the samples watch little tennis (Table 1), and an indicator for any tennis viewing does not moderate the effects (-0.024 , $p = 0.724$, pooled; not tabulated). Scale use is addressed twice: estimates from each participant’s first clip alone, free of within-session comparison, are direction-consistent though imprecise (pooled -0.135 , standard error 0.119; not tabulated), and the compression adjustment of Section 3.5 handles the attraction of re-rendered ratings toward the scale midpoint.

What would separate the bundle is a label experiment: hold the footage fixed and vary only

whether raters are told the performer is a real athlete or a synthetic rendering, crossed with the performer’s stated gender. The identity account predicts that the components follow the claim; the richness account predicts that they follow the footage itself. A sample the size of either of our experiments, concentrated on that contrast rather than spread over five conditions, would be better powered for components of the size we estimate than our own designs were; with the belief elicitation added, it would also separate taste from inaccurate beliefs within whichever branch survives.

5 Conclusion

We re-render professional tennis points so that the same performance can be shown with a different performer. Judgments of identical content then separate cleanly. Swapping the displayed gender of a synthetic performer moves no rating at the 5 percent level. Removing the real performer moves both real-footage gaps. The extra severity attached to women’s unforced errors is removed even when the displayed gender is kept, and what remains is the gap those clips carry on their own. The winner-points gap moves the other way and widens, because real presentation had masked about half of the gap those clips carry.

The first lesson is for measurement. Audit and correspondence designs identify discrimination by randomizing a group label on an otherwise fixed application. Applied to our setting, that design would correctly find the label component small and still miss the larger components, because those move only when the performer does. Where evaluation concerns recorded performances, re-rendering makes the complementary design portable: vary the performer, hold the performance fixed.

The decomposition also speaks to blinding as a policy. Field results on anonymized evaluation are split: on the weight of the evidence, blind auditions advanced women ([Goldin and Rouse, 2000](#)), while anonymized applications at firms that volunteered for the program reduced interviews for the minority candidates they were meant to help, which [Behaghel et al. \(2015\)](#) attribute to firm self-selection and to anonymization blocking the attenuation of negative signals. Blinding can also

run against women: on Israeli matriculation exams, girls scored 0.05 to 0.18 standard deviations higher relative to boys when their own teachers graded them than on anonymously graded state exams covering the same material, in all nine subjects (Lavy, 2008). Our estimates show within one setting why the sign can go either way: real presentation adds severity when women fail and applies a more lenient standard when they succeed, so blinding lifts a penalty and withdraws a subsidy at the same time. The net effect of blinding then depends on the mix of successes and failures being judged, not on the direction of bias alone.

Which feature of real presentation carries the components, the belief that a real woman stands behind the play or the visual richness that only genuine footage carries, is the question our design leaves open, and the label experiment of Section 4.3 is built to answer it. As synthetic renderings of real people become routine in media and evaluation, whether a performer is shown as real stops being fixed background and becomes a design choice. Our results say that choice is not neutral: it decides whether women face extra blame for their failures and lenient standards for their successes, or neither, leaving the gaps the content itself carries.

Ethics and Preregistration

The studies were conducted in accordance with the protocol approved by the Human Subjects Committee of the Faculty of Business, Economics and Informatics, University of Zurich (OEC IRB #2024-060, 18 June 2024). All participants provided informed consent. Both experiments were preregistered in the AEA RCT Registry (AEARCTR-0014001, <https://www.socialscienceregistry.org/trials/14001>).

CRedit Authorship Contribution Statement

Yu Pan: Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing, Funding acquisition. **Carlos Gomez-Gonzalez:** Conceptualization, Methodology, Writing – review & editing. **Helmut M. Dietl:** Con-

ceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

All authors declare no competing interests.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the authors used Claude (Anthropic) in order to improve the language and readability of the text. In creating the video stimuli, the authors used Wonder Studio (Wonder Dynamics), an AI-assisted visual-effects tool, for markerless motion capture and for compositing the author-designed avatar characters into the original footage, as described in Section 2.1. These tools were not used for study design, data analysis, or interpretation. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data Availability

Anonymized participant-level data for both experiments, the experimental materials including the complete set of video stimuli for every display condition, the pre-analysis plan, and the analysis code that generates the results reported here are deposited on the Open Science Framework and are available at <https://osf.io/93fhd/>.

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Appendix A Survey Materials and Instructions

Both experiments used the same survey flow: Prolific advertisement, consent form, Prolific ID entry, eight videos in individually randomized order, and demographic questions. Only the clips and the rating item differed between experiments. The materials below are transcribed verbatim from the administered surveys.

A.1 Prolific Advertisement

- Title: “Watch 8 Tennis Highlights!”
- “You will watch 8 tennis highlights in this study! (No Audio)”
- “The survey is estimated to take approximately 3 minutes!”
- “You will receive a payment of £0.50!”
- Notes: “Please use only Google Chrome for your browser. If you cannot load the videos, please return the survey to save your time.”
- Eligible devices: desktop, tablet.

A.2 Consent Form (Excerpts)

- Study title: “Evaluation of tennis performance.”
- Purpose: “The purpose of this study is to understand how individuals evaluate tennis performance.”
- Procedure: “You will be asked to watch a series of short tennis videos and answer questions about the players’ performance. Your answers will be anonymous, and we will never collect any identifiable information.”

A.3 Video Pages

Each of the eight video pages displayed one clip above the instruction “Please watch the highlight and answer the following question.” The rating item appeared below the clip:

- Experiment 1: “How impressive is the win of this point?”, options displayed from *Very Impressive* (top) to *Very Unimpressive* (bottom);
- Experiment 2: “How severe is this error?”, options displayed from *Very Severe* (top) to *Very Minor* (bottom).

Responses were coded 1–7 so that higher values indicate a more impressive win (Experiment 1) or a more severe error (Experiment 2). Clips were introduced only as tennis highlights: no condition’s clips were described as replays of real points or as animations, and all clips were presented without sound.

A.4 Demographic Questions

- “Were you able to watch all the videos?” (Yes/No)
- “How often do you watch tennis matches or highlights?” (Never; A few times per year; About once a month; Two to three times a month; About once a week; Two to three times a week; Daily)
- “What is the highest level of education you’ve completed?” (five options, from “High school, GED, or less” to “Master’s degree or higher”)
- “What is your household income before taxes?” (eight brackets, from “Less than \$20,000” to “\$200,000 or higher”)

Age, sex, and race were provided by the Prolific platform.

Appendix B Robustness

This appendix reports the robustness battery for the re-rendering effect. Table B.1 varies the specification and the sample and re-estimates the pooled compression-adjusted model with each clip omitted in turn; Figure B.1 reports randomization inference for the baseline specification.

TABLE B.1
ROBUSTNESS OF THE RE-RENDERING EFFECT

	Experiment 1 (Winner Points)	Experiment 2 (Unforced Errors)	Pooled
<i>Panel A: origin gap $\times$ re-rendered</i>			
Baseline specification	−0.099** (0.042)	−0.172*** (0.047)	−0.134*** (0.031)
Straight-line responders retained	−0.095** (0.041)	−0.161*** (0.044)	−0.127*** (0.030)
Experiment 1 participants retained	−0.099** (0.042)	−0.175*** (0.046)	−0.136*** (0.031)
Compression-adjusted	−0.107** (0.043)	−0.102** (0.051)	−0.105*** (0.033)
<i>Panel B: leave-one-clip-out, pooled compression-adjusted estimate</i>			
Range of the 16 estimates	all negative; [−0.137, −0.064]; largest $p = 0.058$		

Note.—Each Panel A cell is the origin gap $\times$ re-rendering interaction from ordinary least squares on the 1–7 ratings, standard errors clustered by participant in parentheses; p -values and stars are z -based. The pooled column adds experiment-by-origin and experiment-by-re-rendering controls. Straight-line responders are the 67 (Experiment 1) and 100 (Experiment 2) participants who gave an identical rating to all eight clips (retained: $N = 1,964$ and $1,933$). The baseline Experiment 2 sample excludes the 25 participants who also took part in Experiment 1 (Section 2.4); the retained row restores them ($N = 1,858$). The compression-adjusted specification adds clip fixed effects and a re-rendering $\times$ clip-extremity control, where clip extremity is the clip’s control-condition mean rating minus the scale midpoint of 4. Panel B re-estimates the pooled compression-adjusted model with each of the 16 clips omitted in turn. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

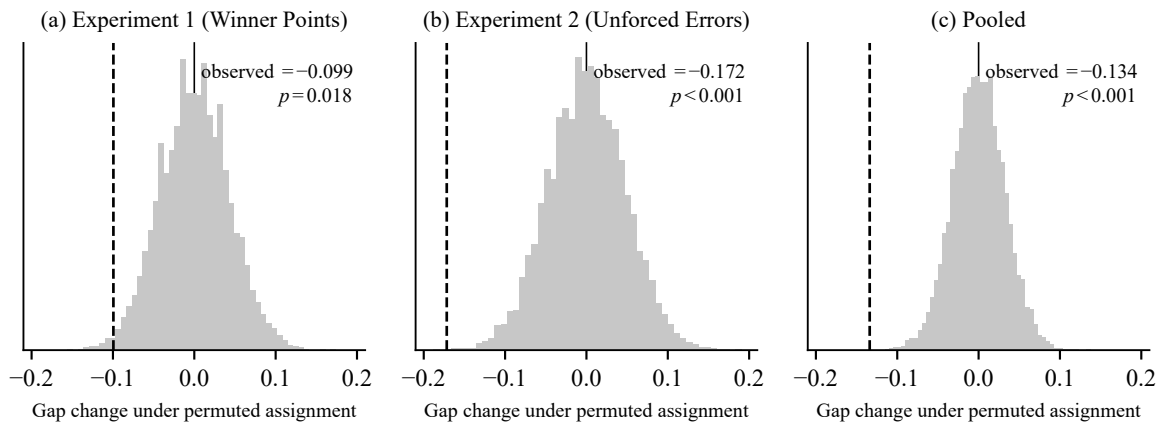


FIGURE B.1

RANDOMIZATION INFERENCE FOR THE RE-RENDERING EFFECT

Note.—Each panel shows the distribution of the origin gap $\times$ re-rendering estimate under 10,000 random permutations of presentation assignment across participants (within experiment in the pooled panel), computed on participant-level gap scores; the dashed line marks the observed estimate. The pooled panel uses the experiment-demeaned regression coefficient. p -values are two-sided.

Appendix C Exploratory Results

This appendix reports the preregistered rater-sex moderations together with the exploratory moderations and category contrasts referenced in Sections 3 and 4 (Table C.1).

TABLE C.1
EXPLORATORY MODERATIONS AND CATEGORY CONTRASTS

<i>Panel A: moderation of the re-rendering effect (three-way interaction)</i>			
	Experiment 1	Experiment 2	Pooled
× Male rater	0.130 (0.083)	0.045 (0.094)	0.095 (0.062)
× Rater age (per 10 years)	0.011 (0.029)	0.072** (0.029)	0.040* (0.020)
<i>Panel B: Experiment 2 severity contrasts on identical content (clip fixed effects)</i>			
	Estimate	Standard error	Participants
All-female avatar vs robot	0.192**	(0.085)	698
All-female vs all-male avatar, female raters	0.255**	(0.113)	349
within-participant implied estimate	−0.012	(0.067)	554
All-female vs all-male avatar, male raters	0.055	(0.113)	391

Note.—Ordinary least squares on the 1–7 ratings; standard errors clustered by participant in parentheses; p -values and stars are z -based. Panel A adds the moderator and all its interactions to the pooled specification of Tables 2 and 3; cells show the three-way interaction, and the pooled column adds experiment-by-origin and experiment-by-re-rendering controls. The rater-sex rows are the preregistered hypotheses H1D and H2D (registered as non-directional); all other rows are exploratory and unadjusted for multiple testing. Panel B compares rated severity of identical content across display conditions in Experiment 2 with clip fixed effects; the within-participant implied estimate applies the Table 4 estimator to female raters. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.